Assessment of Ethanolic Seed Extract of Garcinia Kola on the Brain of Lead Exposed Adult Male Wistar Rat

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ABSTRACT

Garcinia kola (*G. kola*) is a known potential source of antioxidant and anti-inflammatory agents. The present study investigated the effects of ethanolic extract of *G. kola* seed on lead-induced neurotoxicity in adult male Wistar rats. Sixty rats were randomly divided into 4 groups with group A as control. Groups B1, B2 and B3 were untreated lead-exposed (100 ppm), group C1, C2 and C3 were lead-exposed and treated with *G. kola* seed extract (150 mg/kg body weight) or while group D1, D2 and D3 were lead-exposed treated silymarin (150 mg/kg body weight) for 7, 14 and 21 days respectively. Data were analyzed using one-way ANOVA in GraphPad Prism version 10.4.0 at p value < 0.05. *G. kola* treatment significantly decreased brain tissue protein carbonyl by days 7 and 21, MDA by day 14 and GPx by day 21 while brain tissue superoxide dismutase and catalase activities were significantly increased by days 7 and 21 with *G. kola* treatment. This action of *G. kola* was similar with silymarin. The findings indicate that *G. kola* contains essential bioactive compounds with antioxidant properties that help ameliorate the brain tissue Pb-induced oxidative stress.

Keywords: *Garcinia kola,* Oxidative stress, Lead-induced damage, Antioxidants.

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1. INTRODUCTION

Garcinia kola (GK), commonly known as bitter kola, is a dicotyledonous plant widely distributed across the tropical regions of Africa, particularly in Nigeria, Ghana, Benin, Côte d'Ivoire, Cameroon, Sierra Leone, Togo, the Democratic Republic of Congo, Angola, Liberia, and The Gambia [1]. It has long been used in traditional medicine, where it is believed to possess potent purgative, antiparasitic, and antimicrobial properties, as well as therapeutic effects against various infections and liver disorders [2].

Several studies have reported that *G. kola* exhibits antioxidant properties [3] and may serve as a valuable source of anti-inflammatory agents. Notably, its ethanolic extract demonstrates significantly higher antioxidant indices than the aqueous extract. Overall, there is considerable evidence supporting its nutritional and medicinal benefits, with documented biological activities that may be effective against a range of pathogenic and parasitic diseases [4].

Lead, on the other hand, is recognized as one of the most significant environmental pollutants with serious health implications for exposed individuals [5]. Although relatively non-reactive in its natural metallic form, lead has been extensively used in plumbing, car battery manufacturing, gasoline additives, paints, ceramics, and optical technologies [6]. Human exposure occurs primarily through

environmental accumulation, ingestion, inhalation, dermal absorption, and transplacental transfer from mother to fetus.

Once in the body, lead bioaccumulates over time. The most common form is sulfide ore (galena, PbS), which is widely distributed globally. Lead compounds differ in solubility: sulfide, phosphate, carbonate, and oxide forms are practically insoluble in water, whereas acetate and nitrate salts are water-soluble [7].

Considering that *G. kola* exhibits antioxidant properties [3] and biological activities that may be effective against pathogenic and parasitic diseases [4], it is considerate to think of it as a heavy metal chelator. Hence, this study aim to investigates the effects of ethanolic extract of *G. kola* seed on lead-induced neurotoxicity in adult male Wistar rats.

2. MATERIALS AND METHODS

2.1. Experimental animals/ care of laboratory animals: Sixty adults male Wistar rats were obtained from the Animal Holding Unit of the College of Medicine, Ambrose Alli University and transferred to the Department of Anatomy animal house, Ambrose Alli University, Ekpoma, for a two-week acclimatization period. They were maintained on standard rat chow (Bendel Feeds and Flour Mill, Ewu, Nigeria) and provided clean water *ad libitum* in ceramic feeding troughs and drinkers. They were housed in clean plastic cages (45 × 35 × 25 cm) with wire-mesh tops for ventilation, under standard housing conditions consisting of a 12-hour natural light/dark cycle. All animals received standard care, and experimental procedures complied with established guidelines for the care and use of laboratory animals in biomedical research.

2.2. Animal grouping: Group A serves as the control and received standard rat chow and clean tap water only throughout the study. Group B (n = 5) receives lead-contaminated water (100 ppm) for 7 (B1), 14 (B2) and 21 (B3) days. Group C receives lead-contaminated water (100 ppm), and *Garcinia kola* extract (150 mg/kg body weight) for 7 (C1), 14 (C2) and 21 (C3) days and group D recives lead-contaminated water (100 ppm), and silymarin (150 mg/kg body weight) for 7 (D1), 14 (D2) and 21 (D3) days.

2.3 Administration of lead exposed and treatments: Lead (Pb) was administered in drinking water at a concentration of 100 ppm. The Pb solutions were made freely accessible to the animals according to their respective groupings. Oral administration was chosen, as it represents one of the primary routes of heavy metal exposure in both humans and animals.

2.4. Sample collections and analysis: At the end of the study, the animals were sacrificed by cervical decapitation, and the brains were harvested. Brain weight was measured using a digital weighing scale. The harvested brains were then preserved in an ice-filled cooler and transported to the laboratory for further analysis.

2.5 Sample Analysis: Protein carbonyl content was determined using the 2,4-dinitrophenylhydrazine (DNPH) assay as described by Levine et al. [8]. Superoxide dismutase (SOD) activity was measured according to the method of Misra and Fridovich [9]. Catalase (CAT) activity was assessed following the procedure of Cohen et al. [10], while glutathione peroxidase (GPx) activity was determined according to Flohe and Gunzler, [11]. Lipid peroxidation was evaluated by measuring malondialdehyde (MDA) levels using the thiobarbituric acid reactive substances (TBARS) assay, as described by Varshney and Kale [12].

2.6 Statistical Analysis: Data were analyzed using GraphPad Prism statistical software version 10.4.0. Differences between groups were assessed using one-way analysis of variance (ANOVA). Results are presented as mean ± standard deviation (SD). A p value < 0.05 was considered statistically significant..

3. RESULTS

Figure 1 showed the effect of G. kola on brain tissue protein carbonyl (a, b and c) and MDA (d, e, and f) following Pb exposure. Brain tissue protein carbonyl was significantly decreased in the G.kola and silymarin treated groups by day 7 and in the G. kola treated group by day 21 compared to the control. On the other hand, brain tissue MDA level was only significantly decreased in the G. kola treated group compared with the control by day 14.

Figure 2 shows the effect of G. kola on brain tissue antioxidant enzymes following Pb-exposure in rats. Comparatively, brain tissue SOD level was significantly increased in the G. kola and silymarin treated groups compared with the control by days 7 and 21 of threatment (Figure 2a, b and c). On brain CAT level, G. kola treatment resulted in a significant increase by days 7 and 21 while silymarin treatment was only significant by day 21. Brain tissue GPx was however significantly decreased by day 21 in G. kola and silymarin treated groups compared with the control.

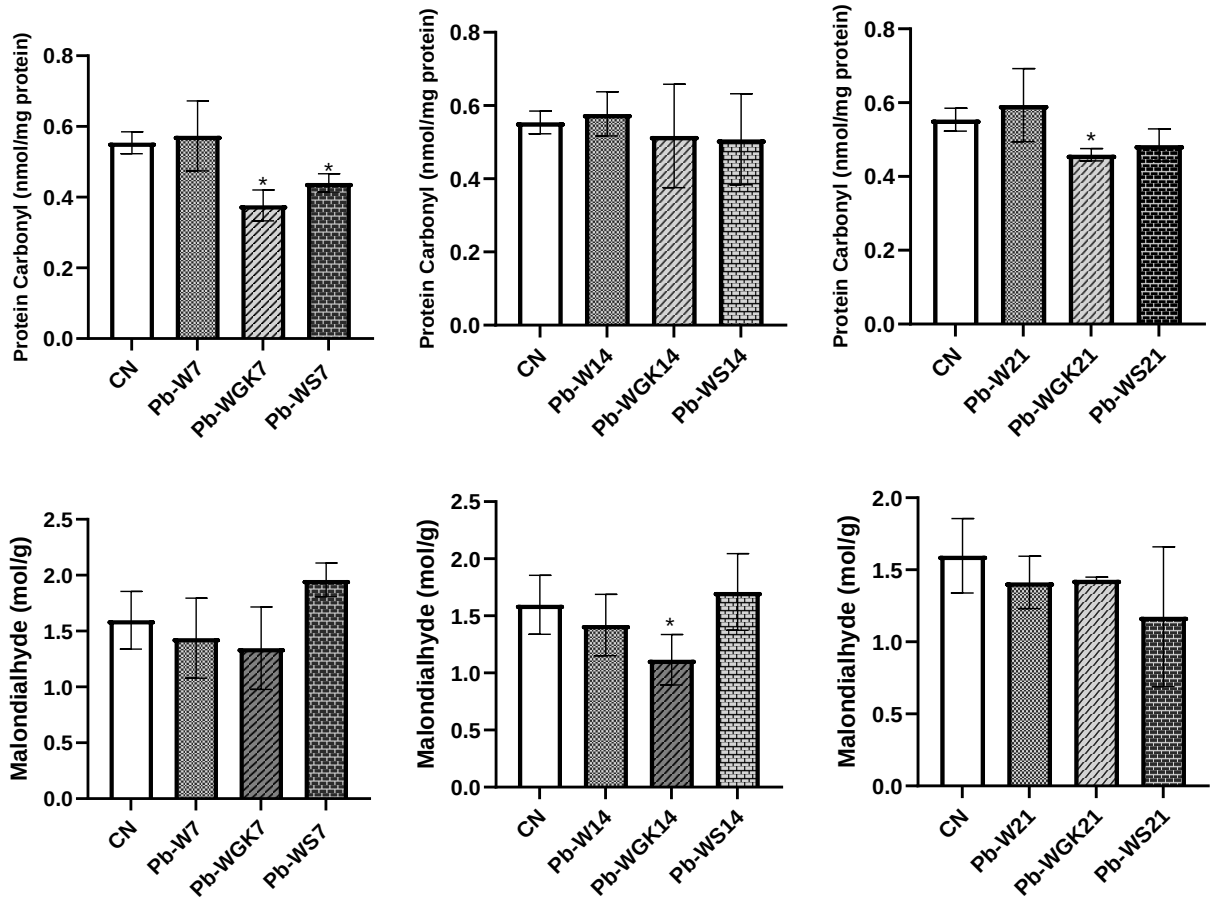


Figure 1. Brain tissue protein carbonyl by 7 (a), 14 (b) and 21 days (c) and MDA by day 7 (d), 14 (e) and 21 (f) following treatment with Garcinia kola post Pb-exposure.
CN=Control; Pb-W7=Group given Lead water only; Pb-WGK7= Group given lead water and treated with *Garcinia kola;* Pb-WS7= Group given lead water and treated with Silymarin

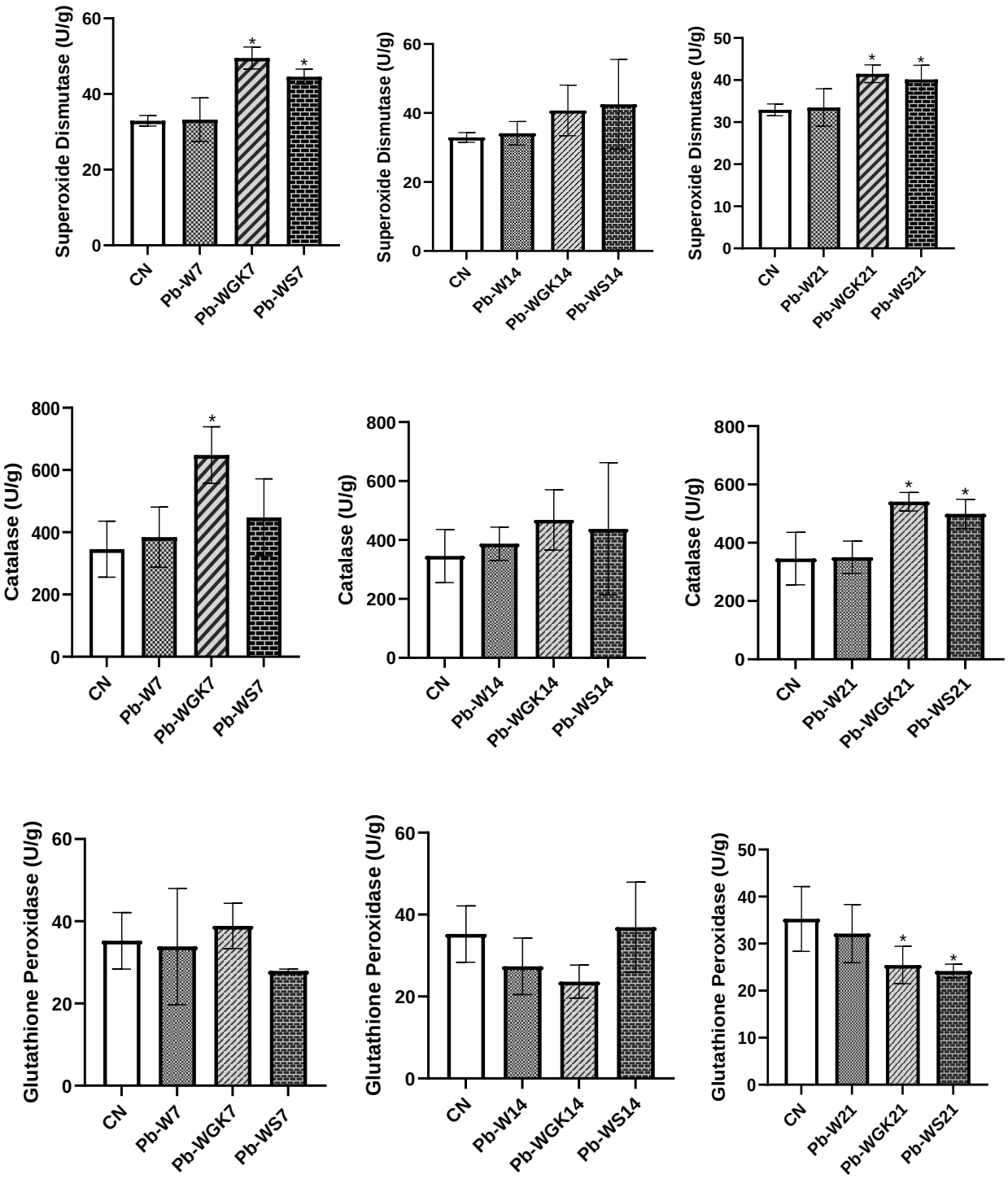


Figure 1. Brain tissue SOD by 7 (a), 14 (b) and 21 days (c), CAT by day 7 (d), 14 (e) and 21 (f) and GPx by day 7 (g), 14 (h) and 21 (i) following treatment with Garcinia kola post Pb-exposure.

CN=Control; Pb-W7=Group given Lead water only; Pb-WGK7= Group given lead water and treated with *Garcinia kola;* Pb-WS7= Group given lead water and treated with Silymarin

4. DISCUSSION

This study evaluated the effects of *G. kola* on lead-induced neurotoxicity in adult male albino Wistar rats by assessing oxidative stress markers and antioxidant status of the brain in rats. the findings showed G. kola extract to significantly mitigate lipid peroxidation that is comparable to the standard drug; silymarin.

Brain tissue protein carbonyl and MDA levels were significantly reduced in groups treated with *G. kola* and silymarin compared with the untreated as well as the control. These findings are consistent with earlier reports by Mazi et al. [13] who demonstrated that kolaviron, a biflavonoid constituent of *G. kola*, enhanced antioxidant defenses by lowering protein carbonyl levels in the brain. Similarly, Dah-Nouvlessounon et al. [14] showed that kolaviron exerted neuroprotective effects against okadaic acid–induced cognitive deficits by reducing protein carbonylation, while John et al. [15] reported cognition-enhancing properties linked to improved antioxidant defense and cholinergic activity. Similar observations were reported by Omotosho et al. [16] have reported kolaviron to reduced MDA levels in rat brain regions following lead exposure. Ishaola et al. [17] also suggested that kolaviron enhanced cholinergic activity and indirectly reduced MDA, while David et al. [18] documented reduced MDA levels and improved neuronal health following administration of *G. kola* ethanol extract.

The reduction of protein carbonyl levels by *G. kola* is of physiological importance, as protein carbonylation is a key marker of protein oxidation that can impair protein structure and function [19]. Lower protein carbonyl levels preserve enzyme activity, maintain cellular functions, and enhance metabolic efficiency, which is especially critical in the brain where oxidative stress is closely associated with neurodegenerative disorders [20, 21].

Treatment with *G. kola* significantly increased SOD and CAT activities compared with control and lead-only groups. This finding align with those of Ali et al. [22], who demonstrated that *G. kola* significantly enhanced SOD and CAT activity, and Adaramoye [23] as well as Sewani-Rusike et al. [24], who reported similar antioxidant-enhancing effects of kolaviron.

In contrast, the effect of *G. kola* on GPx activity was inconsistent and not duration dependent. No significant differences were observed at 7, 14, or 21 days when compared with control or silymarin groups. This observation is supported by Ebana et al. [19], who reported no significant improvement in GPx or SOD activity following *G. kola* treatment. Likewise, Fokou et al. [25] found no statistical difference in glutathione levels between treated and control animals. However, other studies reported contradictory results. Jamila et al. [26] observed higher glutathione concentrations in *Garcinia kola*–treated groups compared to untreated controls, while Jouda et al. [27] demonstrated that kolaviron restored antioxidant enzyme activity and elevated glutathione levels in diabetic rats.

5. CONCLUSION

The findings of this study demonstrate that *Garcinia kola* possesses significant neuroprotective properties against lead-induced oxidative stress in rats. Treatment with *G. kola* markedly reduced protein carbonyl levels, indicating its ability to attenuate protein oxidation and preserve cellular integrity. The extract also enhanced the activities of key antioxidant enzymes, thereby strengthening endogenous defense mechanisms against oxidative stress.

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